XRPL Validator — Daily Engineering Report · 2026-10-11

Period: 2026-10-11 00:00 to 24:00 US Eastern (2026-10-11 04:00 to 2026-10-12 04:00 UTC); ledgers #107,575,702 to #107,598,075. Every count from the ledger and the validator's logs stops at #107,598,075, closed 03:59:52 UTC. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,365 of the 22,374 ledgers the network closed (99.960%); every validation matched the network's validated ledger. Unsigned: 9 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 1 deployment (cycle 169), 1 commit merged into main, 1 engine finding fixed (F420).
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,374
Validations issued	22,365
Coverage (validations / ledgers closed)	99.960%
Validations matching the network's validated ledger hash	22,365 of 22,365
Unsigned during planned restarts	9
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,365
External agreement, data.xrpl.org (UTC day 2026-10-11)	0.99958 (9 missed of 21,449)

Restarts:

- 05:14:04 UTC: deployment of cycle 169; validations resumed at #107,576,862 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,575,702–#107,598,075), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
169	`2053735`	05:14:53 UTC	1,978 passed / 0 failed; port 1,006 /	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0	414 matched, 0 mismatched	#44 from 05:41 UTC: CLEAN (2026-10-12

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
			0	mismatches		05:42 UTC)

Cycle 169 delivered: `4bb4877` docs: fixBatchV1_2 is ported (cycle 168), so task 1.5 is done and the known-gap list starts empty; `6bd1612` fix(probe): an RPC error that does not settle a lookup is a failed fetch, not "absent"; `452f251` fix(ledger): F418 — a permission by name and by number is one permission, so re-filing a delegation changes nothing (Mainnet #107570369 EE825DEED447); `2053735` fix(ledger): F419 — AMMCreate links the AMM, sends the LP tokens, then Amount, then Amount2, as rippled does (Mainnet #107572028 AC9F1F115224)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
17:06 UTC	xrpl-validator	`4c9d0f2`	fix(ledger): F420 — fixCleanup3_4_0's implicit pool authorization reaches the book walk (RippleStateHelpers.cpp:592)

Findings fixed

- **fix(ledger): F420 — fixCleanup3_4_0's implicit pool authorization reaches the book walk (RippleStateHelpers.cpp:592)** (`4c9d0f2` , xrpl-validator) Under fixCleanup3_4_0 rippled's requireAuth treats a pseudo-account (AMMID, VaultID or LoanBrokerID on its root) as authorized for a RequireAuth issuer's IOU when its line lacks the auth bit. BookStep.cpp:768 runs that check per offer owner, an AMM pool included. The engine had the branch in require_auth_ter only, which the book walk never calls: the port's flow::view::is_authorized (book_step.rs) and the model's …

Independence program

No independence record for UTC day 2026-10-11; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,374 (#107,575,702–#107,598,075)
Mean close interval	3.86 s
Longest close interval	9 s (before #107,598,070)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-11 17:06	10-11 17:06	xrpl-validator	`4c9d0f2`	fix(ledger): F420 — fixCleanup3_4_0's implicit pool authorization reaches the book walk (RippleStateHelpers.cpp:592)

Times in UTC. A commit appears on the day it was committed to main or merged into it.